OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2
Examinations 2024-2025

Course Instance Code(s)	4BCT1, 4BS2
Exam(s)	Fourth Computer Science and IT, Fourth Science
Module Code(s)	CT421
Module(s) **Bonded**	Artificial Intelligence No [☑] Yes [ ] List bonds
Paper No. **External Examiner(s)** **Internal Examiner(s)**	1 Dr. R. Trestian Dr. E. Howley *Dr. C O'Riordan

Instructions: Answer 3 questions. All questions are equally weighted

Duration 2 hours
No. of Pages 3
Discipline(s) Computer Science

Requirements:

Remove Paper from exam venue, publish online	No [ ]	Yes[☑]
MCQ Answer sheet	No [☑]	Yes [ ]
Handout	No [☑]	Yes [ ]
Formulae & Log Tables**	No [☑]	Yes [ ]
Cambridge Tables 2nd Edition**	No [☑]	Yes [ ]
Graph Paper*** A4 Graph Paper 1mm 0.1cm Squared (Standard)	No [☑]	Yes [ ]
Other Materials	No [☑]	Yes [ ]
Graphic material in colour	No [☑]	Yes [ ]

End of requirements.
PTO

CT421 Artificial Intelligence

Exam Duration: 2 Hours

Question 1

(a) The game of *tic-tac-toe* is played on a 3x3 grid. Two players take turns placing their markers (X and O) in empty cells. A player wins by getting three of their markers in a row (horizontally, vertically, or diagonally).

i) Draw the first three levels of the game tree and demonstrate how the minimax algorithm would evaluate positions at this depth. Assume player X moves first, and that player X is maximizing. (5)

ii) Explain how alpha-beta pruning could be applied to your game tree from part i). Illustrate your answer with an example of a branch that could be pruned. (5)

(b) For the following three approaches, explain, how they differ in the order that nodes are visited:

Breadth first search
Depth first search
Iterative Deepening

Compare the three techniques under the following headings:
Time complexity
Space complexity
Completeness (10)

(c) Explain briefly the main aspects of Monte Carlo Tree Search. (5)

Question 2

(a) The exploration-exploitation balance is a fundamental challenge in search and optimisation problems. Discuss, with respect to this exploration-exploitation issue:

i. the operators of a genetic algorithm
ii. a hill climbing approach (10)

(b) Consider a resource allocation problem where N different tasks need to be assigned to M processors, where each task has a processing time and memory requirement, and each processor has specific memory constraints and processing capabilities.

Design a genetic algorithm solution for this problem. Your answer should include:

i) Chromosome representation
ii) Fitness function design
iii) Selection mechanism
iv) Crossover and mutation operators (15)

Question 3

(a) Using the prisoner's dilemma as an example game, explain the terms *dominant strategy* and *Nash equilibrium.* (6)

(b) The strategy *tit-for-tat* has been studied in the iterated prisoner's dilemma. Explain the strategy and why it has been considered a successful strategy. How would the strategy fare in the game where moves are subject to noise (i.e. a cooperation taken as a defection and vice versa). (9)

(c) Auctions are frequently used as a mechanism for resource allocation in multi-agent systems. Compare and contrast two different auction types (e.g., English, Dutch) in terms of:

their mechanics
the strategic behaviours they encourage
their efficiency in reaching optimal allocations

(10)

Question 4

(a) Ant Colony Optimisation (ACO) is inspired by the foraging behaviour of ants. Explain the key principles of ACO, including the role of pheromones and their evaporation. Describe how ACO could be applied to find the shortest path between two points in a network. (8)

(b) Compare and contrast symbolic AI and connectionist AI approaches. Using a specific application domain discuss the advantages and limitations of each approach. (8)

(c) Deep learning models are often described as "black box" approaches. Explain what this means and why it can be problematic in real-world applications. Describe two different approaches that can be used to make deep learning models more interpretable. (9)

END